APSK 101 — Foundations of Skeptical Inquiry
3 Credit Hours | Undergraduate Foundation Course
This is a full semester course. Every assignment requires the student to show how they reached a conclusion, not merely give the "right answer."

Unit 1 — Claim or Not a Claim?
Assignment 1: The Claim Audit
For each statement, the student must identify whether it is a fact claim, opinion, prediction, interpretation, value judgment, or unverifiable statement, then explain the choice.
Student work:
"Crime is getting worse."
"New York City has more than eight million residents."
"That movie was terrible."
"Children learn better when they enjoy their teacher."
"This company failed because management was incompetent."
"Artificial intelligence will eventually replace most jobs."
"Everybody knows politicians lie."
"The defendant looked nervous, so he probably knew something."
"This medicine worked for my cousin."
"This is the best way to solve the problem."
For each statement, answer:
A. What exactly is being claimed?
B. What type of claim is it?
C. What evidence would be needed to evaluate it?
D. What information is missing?
E. How confident could you reasonably be right now: low, moderate, or high? Why?

Unit 2 — What Do You Actually Know?
Assignment 2: The Knowledge Inventory
Choose five things you believe are true.
They cannot all be obvious facts such as "water is wet." At least two should involve something you learned from another person, institution, news source, book, website, or social media.
For each belief, complete:
I believe: ________
How did I originally learn this? ________
Have I personally verified it? Yes / No / Partially
What evidence currently supports it? ________
What part am I assuming? ________
Could the original source have been mistaken? How? ________
What would cause me to change my mind? ________
Then write 300–500 words answering:
What is the difference between believing something, knowing something, and being able to demonstrate something?
There is no required philosophical answer. The grade is based on the quality of the reasoning.

Unit 3 — Evidence vs. Assertion
Assignment 3: Build the Evidence Ladder
Students receive this claim:
“Students who use laptops during class learn less.”
They are given six hypothetical pieces of support:
A. A professor says laptops are distracting.
B. Ten students say they concentrate better without laptops.
C. A viral video claims laptops destroy attention spans.
D. A controlled study compares test performance between laptop and non-laptop groups.
E. A school reports that grades increased after banning laptops.
F. Your friend stopped using a laptop and their grade improved.
Student must rank them from strongest to weakest evidence.
But the grade is not based simply on the order.
For every item, explain:
What does this evidence actually establish?
What does it NOT establish?
What alternative explanation could exist?
What additional information would you need?
Then answer:
Could a weaker piece of evidence ever become stronger if more information were provided?
Give an example.

Unit 4 — Burden of Proof
Assignment 4: Who Has to Prove It?
Evaluate these exchanges.
Case 1
Person A: “There is a secret tunnel underneath this building.”
Person B: “I don't believe you.”
Person A: “Prove there isn't.”
Case 2
Person A: “This dietary program improves memory.”
Person B: “What evidence supports that?”
Person A: “Nobody has proven that it doesn't.”
Case 3
Person A: “My employee stole the missing money because nobody else had access.”
For each case:
Who is making the primary claim?
Who carries the burden of proof?
Has that burden been satisfied?
Is absence of contrary evidence being treated as positive evidence?
What evidence would reasonably settle the issue?
Challenge Question
Does the person making a claim always carry the entire burden of proof?

Defend your answer with an example.

Unit 5 — Source Evaluation
Assignment 5: Same Story, Different Sources
Students select one current nonpartisan factual issue being reported by at least three different sources.
They may not simply decide which source they "like."
Create a source table containing:
Who published it?
Who wrote it?
When was it published?
What evidence does it cite?
Does it link to primary material?
Does the headline accurately represent the article?
What facts appear across all three sources?
What claims appear in only one?
What language appears designed to persuade rather than inform?
What remains uncertain after reading all three?
Final question:
If all three sources agree, does that automatically make the claim true? Explain.

Unit 6 — Cognitive Bias
Assignment 6: Catch Yourself
Instead of merely memorizing definitions, students must identify three situations from their own ordinary decision-making where bias could have affected their reasoning.
Choose among concepts such as:
confirmation bias, anchoring, availability bias, motivated reasoning, hindsight bias, framing, or sunk-cost thinking.
For each:
What happened?
What did you initially believe?
What information received more attention because of that belief?
What information may have been ignored?
What would a less biased evaluation have looked like?
Important rule:
A student does not receive extra credit for claiming they overcame the bias.
The point is recognizing that skepticism applies to your own thinking too.

Unit 7 — Correlation, Cause, and Explanation
Assignment 7: What Caused What?
Students analyze:
A city installs additional streetlights. During the following year, reported crime falls 15%.
Answer:
Did the streetlights cause the reduction?

You may answer yes, no, possibly, or insufficient evidence—but you must justify it.
Identify at least five alternative explanations.
Then design a better method for investigating whether the streetlights actually contributed to the decline.
Second scenario:
Students who eat breakfast have higher average test scores than students who skip breakfast.
Explain why this alone does not demonstrate that breakfast caused the difference.

Midterm — Skeptical Analysis
Students receive an unfamiliar claim and have 90 minutes to analyze it.
They must identify:
The exact claim
The burden of proof
Available evidence
Missing evidence
Assumptions
Possible alternative explanations
Source limitations
Their present confidence level
The final answer cannot simply be "true" or "false."
The student must conclude:
Supported / Partially Supported / Insufficient Evidence / Contradicted
and defend the classification.

Unit 8 — Expertise and Authority
Assignment 8: The Expert Problem
Scenario:
A world-famous heart surgeon publicly gives advice about designing earthquake-resistant bridges.
Questions:
Is the surgeon an expert?
Is the surgeon an expert concerning this question?
Does being highly educated increase the reliability of this particular claim?
What qualifications would actually matter?
Could the surgeon still be correct?
If correct, would their medical credentials be the reason the claim is correct?
Then students create their own example of real expertise being incorrectly transferred into another field.
This starts preparing them for the later Doctor/Rocketship Test without teaching the entire Analogical Systems course yet.

Unit 9 — What Would Change Your Mind?
Assignment 9: Falsifiability Exercise
Choose one claim you currently accept.

Write:
Evidence that would increase my confidence: ________
Evidence that would decrease my confidence: ________
Evidence that would cause me to reject the claim: ________
Evidence that sounds impressive but would actually be irrelevant: ________
Then answer:
If you cannot describe any possible evidence that would change your conclusion, are you still investigating the claim?
Minimum 500 words.

Unit 10 — Steelman Before You Attack
Assignment 10: Opposing Position Exercise
Students choose a disputed proposition approved by the instructor.
First, write 500 words defending the position you disagree with.
The defense must be strong enough that someone who actually holds that position would recognize it as fair.
Only afterward may the student critique it.
Then answer:
Did accurately understanding the opposing argument change anything about your original position?
"No" is an acceptable answer—but it must be explained.

Unit 11 — Confidence, Not Certainty
Assignment 11: Confidence Scale
Students evaluate ten claims using:
0–20% — Very Low Confidence
21–40% — Low Confidence
41–60% — Uncertain/Mixed
61–80% — Moderate Confidence
81–95% — High Confidence
96–100% — Exceptional Evidence Required
For every rating they must explain:
Why is your confidence this high but not higher, or this low but not lower?
That forces students away from binary TRUE/FALSE thinking.

FINAL PROJECT — The Skeptical Inquiry Audit
This is the big APSK 101 assignment.
The student selects one substantial real-world claim and conducts a complete investigation.
Their final submission must contain:
The original claim
A precise rewritten version of the claim
Who carries the burden of proof
Primary and secondary sources
Evidence supporting it

Evidence challenging it
Source-quality analysis
Known assumptions
Potential cognitive biases
Alternative explanations
Important missing evidence
What evidence would change the student's conclusion
Final confidence assessment
And the final line must take this form:
“Based on the evidence presently available, I am ___% confident that ______ because ______.”
Then:
“The strongest reason I could be wrong is ______.”
That last sentence is required.